

Notes: Aguirregabiria (RES, 1999)
The Dynamics of Markups and Inventories in Retailing Firms

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1 Big picture

- **Question.** Can the joint dynamics of prices and inventories explain why retail prices often stay unchanged for long periods and yet temporary price reductions (sales promotions) occur frequently?
- **Setting.** The paper builds a dynamic model of a **monopolistically competitive retailer** that jointly chooses **retail prices** and **orders to suppliers**, then estimates it using monthly data for **534 brands** sold by one supermarket chain in the **Basque region of Spain** from **January 1990 to May 1992**.
- **Core challenge.** A standard menu-cost pricing model explains infrequent price changes, but it does *not* easily explain frequent short-lived price cuts. At the same time, a simple inventory model explains lumpy orders, but not nominal price rigidity. The paper's goal is to connect the two.
- **Main theoretical contribution.** The model combines:
 1. an (S, s) **inventory problem** with fixed ordering costs;
 2. a **pricing problem with menu costs**;
 3. **stockout risk**, which makes price and inventories strategic substitutes.

This combination generates cycles in which inventories jump up when orders arrive, markups fall, and then markups rise gradually as inventories are depleted.

- **Why it matters.** The paper argues that an important part of retail markdowns and sales promotions can come from **inventory management**, not only from search frictions, intertemporal price discrimination, or random discounting.
- **Main empirical question.** How much of the frequency of sales promotions and the within-brand variability of retail markups can be explained by fixed ordering costs and the implied inventory cycles?
- **Main empirical results.**
 - Inventories have a strong and robust effect on price decisions, especially on **sales promotions**.
 - Retail markups relative to wholesale cost follow a **cyclical pattern**: they fall when orders are placed and rise between orders.
 - Estimated **fixed ordering costs** are large, around **\$278 per order** in the preferred specification.
 - Estimated **menu costs are asymmetric**: a price increase costs about **\$138.67**, while a price reduction costs about **\$51.46**.
 - In counterfactual simulations, removing fixed ordering costs reduces the monthly frequency of price reductions from **50.8% to 26.8%** and lowers the within-brand standard deviation of markups from **7.3% to 3.8%**.
 - The paper concludes that roughly **half of sales promotions and half of markup variability** in the data are associated with inventory dynamics in the presence of fixed ordering costs.

- **Economic takeaway.** Temporary price cuts can be a rational response to inventory replenishment. When firms face stockout risk and costly orders, a price reduction right after an order arrives can be optimal even if regular nominal prices are otherwise sticky.

2 Data and measurement (key objects)

2.1 The supermarket chain and institutional setting

- The firm is a supermarket chain in the **Basque region of Spain**. During the sample period, the market leader had more than **40%** of the regional market, while this chain had a market share of roughly **3%–4%**, the sixth largest in the region.
- The chain had a **central store** and more than **60 outlets**. Most decisions were centralized: the central store ordered from wholesalers, held inventories, allocated goods to outlets, and decided retail prices.
- Prices were effectively decided at a **weekly** frequency. At the beginning of each week, the central store sent outlets a list of available brands and retail prices. Outlets placed orders to the central store through a computer network, and if inventory was available the goods were delivered within **24 hours**.
- Supplier contracts were negotiated roughly once a year. These contracts specified wholesale prices, discounts, trade promotions, payment terms, and especially how many promotion weeks a brand would receive and how much of promotion costs would be covered by the wholesaler.
- The empirical analysis treats the central store as a retailer that sells directly to consumers. This abstracts from within-chain outlet logistics and assumes the relevant optimization problem is the chain's centralized decision over prices and orders.

2.2 Raw dataset and working sample

- The chain's database contains monthly information from **January 1990 to May 1992** for every brand sold by the company.
- The raw dataset covers **8,742 brands**. For each brand-month it records:
 - total sales,
 - sales under promotion,
 - orders to suppliers,
 - stock at the beginning of the month,
 - wholesale price,
 - retail price under promotion,
 - regular retail price.
- All prices are measured in **pesetas per item**; all quantities are measured in physical units.
- The **working sample** is a balanced panel of **534 brands**, corresponding to **15,486 brand-month observations**.

- Two filters define the working sample:
 1. the central store must actually hold inventories of the brand;
 2. the brand must be sold in *every* month of the 29-month sample.
- These filters are important. Brands without inventories have no inventory problem from the central store’s point of view, and brands sold only for part of the sample would mechanically create long spells of zero orders that have nothing to do with fixed ordering costs.
- The 534 brands are classified into **7 broad product groups**, including health and beauty, cleaning, pharmacy, nonfood items, soft drinks, food, and alcoholic drinks.

2.3 Constructed variables: prices, orders, inventories, and markups

- The data include two monthly price measures:
 - **regular retail price**;
 - **price under promotion**.

Both are monthly averages, defined as nominal sales value divided by sales in physical units.

- Because regular retail prices are sticky, the monthly averages are informative about actual nominal prices and price changes.
- The paper constructs three price-related objects used in estimation:
 1. the retail price during the month, used to compute the markup;
 2. the retail price at the **beginning of the month**, denoted $p^{[b]}$, used to construct the state variable b ;
 3. a categorical nominal price-change variable $I^{\Delta P} \in \{-1, 0, 1\}$ indicating a reduction, no change, or increase.
- To build these variables, the paper assumes that sales promotions occur roughly in the middle of the month and that regular retail prices change at most once per month.
- The paper defines the retail markup as

$$m_{it} = \ln(P_{it}) - \ln(C_{it}),$$

where P_{it} is the retail price and C_{it} the wholesale price. So the paper studies markup variation relative to **wholesale cost**.

2.4 Key descriptive facts

- Orders are fairly frequent but still lumpy:
 - the mean probability of a positive order in a month is **76.60%**;
 - the mean duration between two positive orders is **1.38 months** (about **5.5 weeks**).
- Markups differ sharply between regular pricing and promotions:

- mean regular-price markup: **20.50%**;
- mean promotion-price markup: **5.10%**;
- mean retail markup overall: **16.68%**.
- Regular retail prices are sticky:
 - the mean frequency of regular-price changes is **46.60%**;
 - the mean duration of a regular-price quote is **2.18 months**;
 - for the median brand, the duration is **1.80 months**.
- Wholesale prices are even stickier:
 - mean frequency of wholesale-price changes is **34.63%**;
 - mean duration is **3.46 months**.
- Sales promotions are very common:
 - the mean frequency of months with promotions is **42.85%**;
 - sales under promotion account for **21.90%** of total sales.
- Figure 5 shows typical brand-level time series in which large orders coincide with large drops in average retail prices. The paper explicitly notes that this correlation alone is not causal evidence, because low prices also stimulate sales and hence future orders.

2.5 Stationarity and sample properties

- The structural estimation requires stationarity of the observable state variables: inventories, markups at the beginning of the month, and real wholesale prices.
- Brand-level Dickey–Fuller tests strongly support stationarity for markups, inventories, and orders. For **61 brands**, real wholesale prices show nonstationarity, but using detrended wholesale prices instead of levels produces negligible differences in the estimates.
- The paper also abstracts from strategic interactions across brands inside the same chain and treats the decision problem as **separable across brands**. This is a deliberate simplification, since the chain actually sells thousands of products.

3 Models and empirical specifications (all equations + interpretation)

3.1 One-period profit, sales, and demand

At every period t , the retailer chooses the retail price and the order quantity to maximize discounted expected profits. The expected one-period real profit is

$$\pi_t = \exp\{p_t\}y_t^e - \exp\{c_t\}q_t - \alpha s_t - \eta^q \mathbf{1}(q_t > 0) - \eta^P \mathbf{1}(P_t \neq P_{t-1}),$$

where:

- p_t is the log real retail price,
- P_t is the log nominal retail price,
- c_t is the log real wholesale price,
- q_t is the order quantity,
- s_t is inventory at the beginning of the period,
- α is the unit storage cost,
- η^q is the lump-sum ordering cost,
- η^P is the lump-sum nominal price adjustment cost.

Actual sales are constrained by inventories:

$$y_t = \min\{y_t^*, s_t + q_t\}.$$

Demand is isoelastic:

$$y_t^* = \exp\{a_t + \gamma_1 p_t + \gamma_2 \tilde{p}_t\},$$

where a_t is a demand shock and $\tilde{p}_t$ is the average price at rival firms. Rewriting demand in terms of the firm's markup $m_t = p_t - c_t$ gives

$$y_t^* = \exp\{a_t + \omega_t + \gamma_1 m_t + (\gamma_1 + \gamma_2)c_t\},$$

with $\omega_t = \gamma_2(\tilde{p}_t - c_t)$ representing rivals' average markup.

Interpretation. The key nonstandard feature is the stockout constraint. Because expected sales depend on the ratio of available inventory to expected demand, a higher price can substitute for lower inventories by reducing stockout risk. This is what links inventory dynamics to markup dynamics.

3.2 Timing, state variables, and Bellman equation

The model assumes:

- rivals' average markup ω_t follows an exogenous first-order Markov process;
- real wholesale price c_t follows an exogenous first-order Markov process;
- the consumers' demand shock a_t is i.i.d.;
- aggregate inflation is constant at rate ρ .

At the beginning of period t , the retailer observes inventories, wholesale price, the previous nominal retail price, and rivals' average markup. The state vector is

$$k_t = (\omega_t, c_t, s_t, b_t)', \quad b_t = P_{t-1} - C_t,$$

where b_t is the markup the retailer would have *if it kept the nominal price unchanged* in period t .

Inventory evolves as

$$s_{t+1} = s_t + q_t - y_t = \max\{0, s_t + q_t - y_t^*\},$$

and the beginning-of-period markup next period evolves as

$$b_{t+1} = m_t - \rho - (c_{t+1} - c_t).$$

The Bellman equation is

$$V(k) = \max_{\{m, q \geq 0\}} \left\{ \pi(k, m, q) + \beta \int V(\omega', c', m - \rho - [c' - c], s') f_a(da') f_c(dc'; c) f_\omega(d\omega'; \omega) \right\}.$$

Interpretation. The relevant dynamics come from three sources of uncertainty: current demand, future wholesale prices, and future rival markups. The retailer chooses a current price and order knowing it cannot revise them until next period.

3.3 Optimal decision rule without menu costs

When menu costs are zero, the model still implies a nontrivial joint pricing-inventory rule. The optimal policy is:

$$q^*(s) = \begin{cases} z^* - s, & \text{if } s < z^L, \\ 0, & \text{if } s \geq z^L, \end{cases} \quad m^*(s) = \begin{cases} m^*, & \text{if } s < z^L, \\ \tilde{m}(s), & \text{if } s \geq z^L, \end{cases}$$

where z^* is the optimal target inventory level absent adjustment frictions, $z^L < z^*$ is the reorder threshold, and $\tilde{m}(s)$ is a decreasing function of inventories.

Interpretation. This is an (S, s) **inventory rule** plus an endogenous markup rule. Between orders, inventories fall and markups rise because the probability of stockout increases and demand becomes less elastic. When an order is placed, inventories jump up and the markup drops.

3.4 Optimal decision rule with menu costs

With menu costs, the state space (b, s) is partitioned into four regions corresponding to whether the retailer changes price and whether it orders. The decision rules become

$$q^*(b, s) = \begin{cases} z^* - s, & \text{if } (b, s) \in T_{11}, \\ \tilde{z}(b) - s, & \text{if } (b, s) \in T_{01}, \\ 0, & \text{if } (b, s) \in T_{00} \cup T_{10}, \end{cases}$$

$$m^*(b, s) = \begin{cases} m^*, & \text{if } (b, s) \in T_{11}, \\ \tilde{m}(s), & \text{if } (b, s) \in T_{10}, \\ b, & \text{if } (b, s) \in T_{00} \cup T_{01}. \end{cases}$$

The four regions are:

- T_{00} : no price change and zero orders,
- T_{01} : no price change and positive orders,
- T_{10} : price change and zero orders,

- T_{11} : price change and positive orders.

Interpretation. Menu costs create an **inaction region**. If inventories are high and the gap between the inherited markup b and the desired markup is small, the retailer does nothing. Once inventories fall enough, the retailer either changes the price, places an order, or both. If ordering costs are large relative to menu costs, a sizable temporary price cut can occur precisely when orders are placed.

3.5 What the theory predicts for promotions and markups

The model predicts:

- long periods with no nominal price changes,
- short periods with very low prices right after replenishment,
- positive duration dependence in sales promotions,
- cyclical markups that track inventory cycles,
- possible cross-sectional markup dispersion even when menu costs are small or zero, because firms may be at different points of their inventory cycle.

Interpretation. The retailer's low-price episodes are not random. They are state dependent and tied to the inventory position and the timing of orders.

3.6 Reduced-form empirical tests of the model

The paper first estimates fixed-effects probit models for:

- the discrete order decision $q > 0$,
- reductions and increases in regular retail prices,
- sales promotions.

The explanatory variables are the observable state variables $\ln(s)$, b , and c , plus brand fixed effects and month dummies. Some specifications also include lagged sales $\ln(y_{t-1})$ to capture possible autocorrelation in demand shocks.

A separate descriptive specification studies markup dynamics within an ordering spell. If $m_{r(i),j}$ is the markup j months after an order in spell $r(i)$ and $\tau_{r(i)}$ is the spell length, the paper defines

$$\beta_{\tau,j} = \mathbb{E}[m_{r(i),j} - m_{r(i),0} \mid \tau_{r(i)} = \tau].$$

Interpretation. If markups rise between orders, then $\beta_{\tau,j}$ should increase with j , and the increment should get smaller as the next order approaches. This is exactly what the model predicts when ordering costs are important.

3.7 Structural estimation: optimal discrete choice representation

The structural parameters fall into three groups:

- demand parameters (γ_1, γ_2) ,
- transition parameters for the state variables,
- storage and lump-sum cost parameters (α, η^q, η^p) .

The paper emphasizes that lump-sum costs do *not* appear explicitly in Euler equations, so identifying them requires exploiting the **optimal discrete choice**.

With asymmetric menu costs, there are six discrete alternatives: two order choices ($q = 0$ or $q > 0$) times three price-change outcomes (price reduction, no change, price increase). If d indexes these alternatives, the conditional choice value is

$$V^d(k; \theta) = \pi(k, m^d(k; \theta), q^d(k; \theta); \theta_\pi) + \beta \mathbb{E} \left[\max_j V^j(k'; \theta) \mid k, d; \theta \right].$$

The chosen discrete action solves

$$d^*(k; \theta) = \operatorname{argmax}_{j \in \{1, \dots, 6\}} V^j(k; \theta).$$

3.8 Conditional independence and the sequential estimator

The key econometric simplification is the conditional independence assumption:

$$pdf(x', \varepsilon' \mid x, \varepsilon, d) = pdf(\varepsilon' \mid x') pdf(x' \mid x, d),$$

where x denotes the observable state vector and ε is the vector of latent payoff shocks associated with the discrete alternatives.

Under this assumption and a multiplicative separability condition for current profits, the optimal discrete choice can be written as

$$d^*(k_{it}; \theta) = \operatorname{argmax}_{j \in \{1, \dots, 6\}} \left\{ \Pi^j(x_{it})' \mu(\theta_\pi) + W_{it}^j \lambda(\theta_\pi) + \varepsilon_{it}^j \right\},$$

where:

- $\Pi^j(x_{it})$ summarizes current expected profit components,
- W_{it}^j summarizes discounted future-value components,
- $\mu(\theta_\pi)$ and $\lambda(\theta_\pi)$ collect structural parameters.

The model is then estimated in two stages:

1. **Stage 1:** nonparametrically estimate conditional choice probabilities, state transitions, and expected current-profit components;
2. **Stage 2:** plug those estimates into a GMM discrete-choice problem using moment conditions of the form

$$\mathbb{E} \left[Z_{it} (\mathbf{1}(d_{it} = d) - p^{(d)}(\Pi_{it}, \widehat{W}_{it}; \theta_\pi)) \right] = 0.$$

Interpretation. This sequential method avoids solving the full dynamic programming problem inside every iteration of estimation. That is the computational advantage of the Hotz–Miller style approach used in the paper.

3.9 Selection correction in first-stage profit estimation

Estimating terms like $\mathbb{E}(q_{it} \mid x_{it}, d)$ or $\mathbb{E}(e^{m_{it}} y_{it} \mid x_{it}, d)$ is not straightforward because actual choices depend on unobserved shocks, creating sample selection bias.

The paper uses model-implied exclusion restrictions. For example, in some regimes $q_t + s_t$ does not depend on all state variables. These restrictions are used together with a nonparametric Ahn–Powell selection correction.

Interpretation. The structure of the dynamic model does more than motivate counterfactuals; it also generates the exclusion restrictions needed to recover expected profits under hypothetical discrete choices.

3.10 Counterfactual simulation design

Using the preferred specification, the paper solves the model numerically and simulates 1,000 replications under four scenarios:

1. baseline model with fixed ordering costs, asymmetric menu costs, and demand-shock variation;
2. zero fixed ordering costs;
3. symmetric menu costs;
4. constant competitors' markups.

The simulations use parametric demand estimates from the appendix rather than the semireduced-form sales equation so that the experiments are not vulnerable to the Lucas critique.

Interpretation. These counterfactuals isolate how much price dynamics are driven by ordering costs, by asymmetric menu costs, and by exogenous demand variation.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the naive approach

- Simply observing that orders and temporary price cuts happen at the same time does *not* identify the mechanism. Low prices can themselves increase sales, reduce inventories, and create the need for new orders.
- A pure menu-cost model can generate infrequent price changes, but it struggles with the frequency and timing of promotions.
- Standard Euler-equation approaches are not enough because the parameters of interest – especially fixed ordering costs and menu costs – do not enter those equations directly.

4.2 What identifies the cost parameters

- The structural identification relies on **optimal discrete choice**. Different combinations of price changes and order decisions occur with different probabilities depending on inventories, inherited markups, and wholesale prices.

- Fixed ordering costs are identified from the lumpy ordering pattern and from the fact that markup dynamics move systematically between orders.
- Menu costs are identified from the frequency and asymmetry of nominal price increases and decreases, conditional on the state variables.

4.3 Exclusion restrictions and selection correction

- The paper’s theoretical decision rules imply that in some discrete-choice regimes the optimal continuous decisions do not depend on all observed states.
- These exclusion restrictions are used to estimate current profit components under hypothetical choices, while correcting for selection bias using the Ahn–Powell method.
- The paper also checks the same restrictions in a parametric Heckman-style framework and reports that they are not rejected.

4.4 Conditional independence and spurious state dependence

- The conditional independence assumption is crucial because it makes future-value terms depend only on observable states, not on the entire history of unobservables.
- The paper tests this assumption by including lagged sales in the conditional-choice profit functions. Lagged sales are significant, which suggests some autocorrelation in demand shocks.
- But the Durbin–Hausman–Wu tests show that including lagged sales produces negligible changes in the main structural estimates. So mild misspecification of demand-shock autocorrelation does not seem to drive the central results.

4.5 Instruments and overidentifying restrictions

- The GMM estimation uses as instruments current and lagged state variables up to lag 3:

$$Z_{it} = \{c_{i,t-j}, \ln s_{i,t-j}, b_{i,t-j} : j = 0, 1, 2, 3\}.$$

- The Hansen tests of overidentifying restrictions have p -values above 0.2 in all specifications and around **0.51** in the preferred asymmetric-menu-cost specification, so the moment conditions are not rejected.

4.6 Empirical strengths and limitations

- **Strengths.**
 - very rich monthly brand-level data with prices, sales, stocks, and orders;
 - direct observation of promotions and regular prices;
 - dynamic model tightly linked to the institutional details of supermarket replenishment.
- **Limitations.**

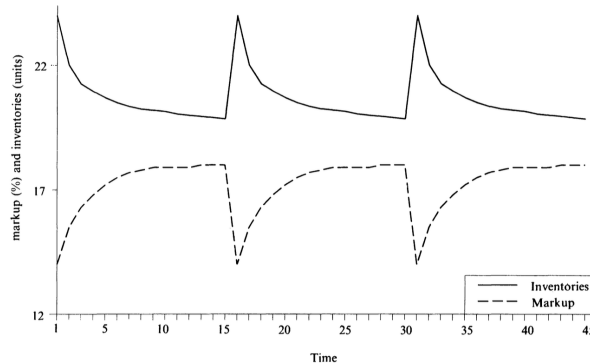


FIGURE 1
Time series of markup and inventories (without menu costs)

- only one supermarket chain is observed, so the paper cannot quantify cross-store or cross-firm markup dispersion in a market;
- decisions across different brands are treated as separable;
- wholesale prices are treated as exogenous and the retailer is price taking in the wholesale market.

5 Tables and figures

5.1 Figure 1: markup and inventories without menu costs

Read:

- Figure 1 is the cleanest picture of the paper’s core mechanism. Inventories follow a sawtooth pattern because orders arrive infrequently, while markups rise between orders and drop when new inventory arrives.
- Even *without* menu costs, fixed ordering costs plus stockout risk are enough to generate cyclical markups and cross-sectional markup dispersion.
- The figure’s message is that inventory cycles are a real source of price and markup dynamics, independent of nominal rigidity.

5.2 Figure 2 and Figure 3: menu costs create an inaction region

Read:

- Figure 2 shows the partition of the (b, s) state space into regions for price changes and orders. The central inaction region T_{00} corresponds to high inventories and a small gap between the inherited markup and the desired markup.
- Figure 3 shows the implied time path with menu costs: inventories still follow an (S, s) -type cycle, but now markup changes are step-like because nominal prices are changed infrequently.
- The important economic point is that **long spells of unchanged regular prices** can coexist with **short low-price episodes** when inventories are replenished.

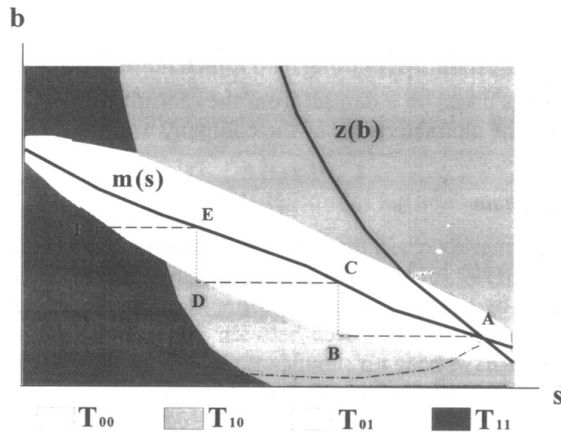


FIGURE 2
Optimal decision rule (model with menu costs)

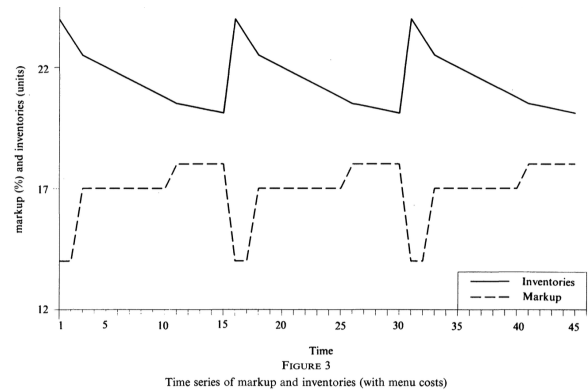


FIGURE 3
Time series of markup and inventories (with menu costs)

5.3 Figure 5 and Table 2: descriptive facts from the data

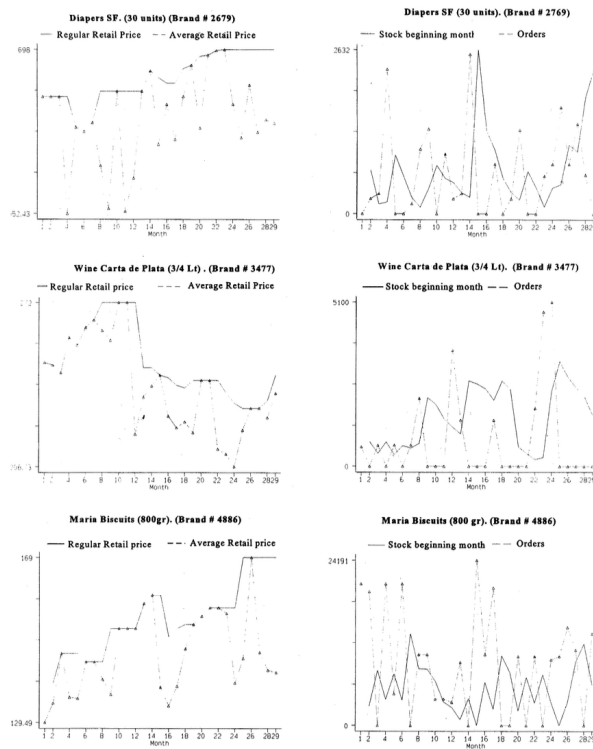


FIGURE 5
Time series for prices, inventories, and orders

TABLE 2

Descriptive statistics: Distribution of brand-specific averages 534 brands. January 1990–May 1992 (29 months)

		Mean	Petile. 10	Petile. 25	Median	Petile. 75	Petile. 90
Orders	(1) Frequency $q > 0$ (%)	76.60	48.28	65.52	79.31	89.66	96.55
	(2) Duration (in months)	1.38	1.04	1.08	1.22	1.44	2.00
Markups	(3) Regular retail price (%)	20.50	9.46	15.69	19.95	24.77	30.31
	(4) Price under promotion (%)	5.10	-0.31	1.82	4.36	7.65	11.43
	(5) Retail price (%)	16.68	5.70	12.12	16.65	21.76	27.14
Regular Retail Price	(6) Freq. price changes (%)	46.60	17.86	28.57	46.43	60.71	78.57
	(7) Duration (in months)	2.18	1.18	1.44	1.80	2.56	3.60
Wholesale Price	(8) Freq. price changes (%)	34.63	10.71	21.43	34.14	42.86	50.00
	(9) Duration (in months)	3.46	1.82	2.10	2.86	3.80	6.00
Sales	(10) Freq. prom. > 0 (%)	42.85	6.90	17.24	37.93	68.97	86.21
	(11) Ratio prom./total sales (%)	21.90	1.59	7.43	18.36	33.57	47.40
Retail Price	(12) Freq. price changes (%)	68.95	35.71	53.57	71.43	85.71	96.43
	(13) Duration (in months)	1.58	1.04	1.12	1.35	1.71	2.27

^a All the statistics in this table have been obtained using the 534 brand-specific means of the variables. Therefore, these statistics refer to the between-brands distribution of the variables.

^b Row 2 presents the average duration between two positive orders.

^c Rows 7, 9, and 13 present the average durations between two nominal price changes.

^d In order to avoid right-censoring bias due to incomplete duration spells, the average durations have been obtained without including those spells starting during the last 5 months.

^e The ratio in row 11 is equal to "sales under promotions (in phys. units)/total sales (in phys. units)".

Read:

- Figure 5 shows actual brand-level series in which average retail prices tend to be lowest when orders are large.
- Table 2 quantifies the basic stylized facts:

- orders are placed every **1.38 months** on average;
 - sales promotions occur in **42.85%** of months;
 - promotions account for **21.90%** of sales;
 - regular-price quotes last **2.18 months** on average, longer than retail prices overall.
- The big descriptive puzzle is already visible here: nominal rigidity in regular prices coexists with frequent promotions.

5.4 Table 3: reduced-form probits for orders, price changes, and promotions

Read:

- The stock at the beginning of the month is the most important predictor of the order decision: in columns (1) and (2), the coefficient on $\ln(s)$ is about -0.91 .
- For regular retail price changes, the inherited markup b is the dominant state variable:
 - in the price-reduction probits, the coefficient on b is about **10.6 to 10.7**;
 - in the price-increase probits, the coefficient on b is about -11.7 .
- Inventories matter much more for promotions than for regular price reductions. The coefficient on $\ln(s)$ in the promotion probit is around **0.109** or **0.088**, much larger than in the regular-price-reduction probit.
- Interpretation: large stocks make promotions more likely, while falling stocks make price increases more likely. This is exactly the inventory channel in the model.

TABLE 3
Fixed-Effects Probit Models for Discrete Choices (Standard errors in parentheses)

	Orders $q > 0$		Regular retail price $\Delta P_{Reg} = -1$ $\Delta P_{Reg} = +1$				Sales promotions $\Delta SP = 1$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(s)$	-0.911 (0.020)	-0.927 (0.020)	0.026 (0.009)	0.018 (0.009)	-0.044 (0.008)	-0.044 (0.008)	0.109 (0.009)	0.088 (0.009)
b	-0.059 (0.243)	0.028 (0.244)	10.585 (0.275)	10.725 (0.276)	-11.728 (0.269)	-11.723 (0.270)	2.309 (0.213)	2.839 (0.219)
c	-0.183 (0.228)	-0.059 (0.232)	0.668 (0.206)	0.840 (0.208)	-0.942 (0.199)	-0.934 (0.201)	0.390 (0.206)	0.377 (0.212)
$\ln y(-1)$		0.084 (0.024)		0.117 (0.022)		0.006 (0.021)		0.384 (0.021)
Log-lik.	-4.993	-4.985	-6.463	-6.448	-7.336	-7.336	-6.668	-6.498
Pseudo R^2 ^a	0.366	0.367	0.232	0.234	0.229	0.229	0.303	0.321
Number obs. ^b	14,056	14,056	14,868	14,868	14,952	14,952	13,888	13,888

^a Pseudo- $R^2 = 1 - \text{RSS}/\text{TSS}$, where $\text{RSS} = \sum_{i=1}^N \sum_{j=1}^J (D_{ij} - \hat{D}_{ij})^2$; $\text{TSS} = \sum_{i=1}^N \sum_{j=1}^J (D_{ij} - \bar{D}_j)^2$; $\bar{D}_j$ is the dummy dependent variable; and $\hat{D}_{ij}$ is the predicted index, between 0 and 1.

^b In these estimations we have not used those brands without enough variability in the dependent variable, i.e. with histories of the dependent variable that contain only 1 zero or only 1 one. This explains the different number of observations for each of the estimations.

TABLE 4
Markup behaviour between two orders

	Estimate of $\beta_{\tau,j}$ (in % points)	s.e.
$\tau = 1$		
After 1 month ($j = 1$)	-0.150	0.100
$\tau = 2$		
After 1 month ($j = 1$)	3.209	0.268
After 2 months ($j = 2$)	0.781	0.268
$\tau = 3$		
After 1 month ($j = 1$)	2.437	0.497
After 2 months ($j = 2$)	3.304	0.497
After 3 months ($j = 3$)	0.313	0.497
$\tau = 4$		
After 1 month ($j = 1$)	3.604	0.782
After 2 months ($j = 2$)	4.576	0.782
After 3 months ($j = 3$)	4.467	0.782
After 4 months ($j = 4$)	1.561	0.782

5.5 Table 4: markup behaviour between two orders

Read:

- Table 4 computes the average increase in markups as time passes within an ordering spell.

- For spells of length 2 months, the markup is about **3.209 percentage points** higher one month after an order.
- For spells of length 4 months, the markup is about **3.604** points higher after one month, **4.576** after two months, and **4.467** after three months.
- These are economically important numbers because the average within-brand standard deviation of markups is only about **5.61 percentage points**.
- So a substantial share of markup variation comes from movement *within* the order cycle.

5.6 Table 5: structural parameter estimates

Read:

- The preferred specification is the asymmetric-menu-cost model with lagged sales controls (column 4).
- The key estimates are:
 - unit storage cost $\alpha = 24.09$,
 - fixed ordering cost $\eta^q = 278.11$,
 - menu cost for price reductions $\eta^{p(-)} = 51.46$,
 - menu cost for price increases $\eta^{p(+)} = 138.67$,
 - discrete-choice shock scale $\sigma_\varepsilon = 0.097$.
- The asymmetry in menu costs is striking: a price reduction costs only about **one third** as much as a price increase.
- The preferred model fits the data substantially better than the symmetric-menu-cost model. The goodness-of-fit p -value is about **0.394**, and the Hansen test has p -value **0.510**.
- The paper interprets the asymmetry as plausible because wholesalers often reimburse a substantial part of promotion-related costs.

TABLE 5
Structural parameters estimates GMM Estimation of discrete choice model 534 brands.
January 1990–May 1992 (29 months)

	Symmetric menu costs		Asymmetric menu costs	
	(1) Estimate (s.e.)	(2) Estimate (s.e.)	(3) Estimate (s.e.)	(4) Estimate (s.e.)
α	33.44 (6.39)	21.06 (6.69)	36.99 (6.23)	24.09 (6.59)
η^q	319.25 (24.76)	311.73 (24.97)	282.26 (21.94)	278.11 (21.40)
η^p	68.55 (25.05)	72.62 (26.24)		
$\eta^{p(-)}$			49.72 (19.87)	51.46 (19.42)
$\eta^{p(+)}$			138.94 (18.05)	138.67 (18.71)
σ_ε	0.145 (0.017)	0.110 (0.020)	0.105 (0.016)	0.097 (0.011)
$\ln y_{i,t-1} (d=1)$		0.303 (0.064)		0.261 (0.087)
$\ln y_{i,t-1} (d=3)$		−0.067 (0.027)		−0.055 (0.024)
$\ln y_{i,t-1} (d=4)$		0.398 (0.162)		0.242 (0.089)
$\ln y_{i,t-1} (d=5)$		0.024 (0.089)		0.052 (0.084)
$\ln y_{i,t-1} (d=6)$		−0.033 (0.047)		−0.021 (0.056)
D-H-W test (p -value)		2.54 (0.469)		3.13 (0.535)
Hansen test (p -value)		64.13 (0.213)	53.87 (0.365)	54.86 (0.480)
Goodness of fit (p -value)		18,091 (0.196)	18,078 (0.201)	17,973 (0.388)
Observations		13,350	13,350	13,350

^a Instrumental variables: $\{c_{i,t-j}, \ln s_{i,t-j}, h_{i,t-j}\}$ for $j = 0, 1, 2, 3$

^b The number of moment conditions is 60, i.e. 12 instruments times 5 equations.

^c Goodness-of-fit test: $\chi^2 = NT \sum_{m=1}^M \sum_{d=1}^D [\hat{p}^d(x^m) - p^d(x^m, \hat{\theta})]^2 / \hat{p}^d(x^m)$, where $M = 3584$ (the number of cells in the discretized state space), $\hat{p}^d(x^m)$ and $p^d(x^m, \hat{\theta})$ are the kernel estimate and the structural estimate, respectively, of the conditional choice probabilities. Under the H_0 , χ^2 is a χ^2 with $5M-K$ degrees of freedom, where K is the dimension of θ .

TABLE 6
Menu costs and fixed ordering costs

Fixed ordering costs				
	Fixed OC per order	Fixed OC/ sales	Fixed OC/ items sold	Fixed OC/ variable OC
	\$278.11	3.15%	\$0.0365	3.35%
Menu costs				
	MC per price change and per store	MC/sales	MC/items sold	
Our study	Total	0.70%	\$0.0081	
	$\Delta P > 0$	0.31%	\$0.0035	
Levy <i>et al.</i>	$\Delta P < 0$	0.39%	\$0.0046	
	(a)	0.70%	\$0.0119	
Slade	(b)	0.72%	\$0.0123	
			\$2.70	

^a Supermarket chains where price tags are placed on the shelves but not on each individual item.

^b Supermarket chains where a separate price tag is placed on each item (in addition to the shelf price tag).

TABLE 7
Counterfactual experiments

	Scenario 1 $\eta_a = 278.1$ $\eta^{(-)} = 51.5$ $\sigma_z = 0.306$	Scenario 2 $\eta_a = 0.0$ $\eta^{(-)} = 51.5$ $\sigma_z = 0.306$	Scenario 3 $\eta_a = 278.1$ $\eta^{(-)} = 138.7$ $\sigma_z = 0.306$	Scenario 4 $\eta_a = 278.1$ $\eta^{(-)} = 51.5$ $\sigma_z = 0.0$
Frequency of positive orders (%)	70.1	100.0	64.2	69.7
Frequency of negative ΔP (%)	50.8	26.8	48.5	44.4
Frequency of positive ΔP (%)	29.6	10.9	19.4	14.2
Std. dev. of markup (%)	7.3	3.8	5.6	4.1

5.7 Table 6: fixed ordering costs versus menu costs

Read:

- Fixed ordering costs are economically large:
 - **\$278.11 per order**,
 - about **3.15%** of sales,
 - about **3.35%** of variable ordering costs,
 - roughly **3.65 cents per item sold**.
- Total menu costs are much smaller in aggregate, about **0.70%** of sales.
- Converting chain-level estimates to a per-store basis gives menu costs of roughly **\$2.23** for a price increase and **\$0.83** for a price reduction.
- Compared with previous studies, these menu-cost magnitudes are broadly similar, while the ordering-cost estimate is a new contribution of the paper.

5.8 Table 7: counterfactual experiments

Read:

- The baseline simulation (scenario 1) produces:
 - positive orders in **70.1%** of months,
 - negative price changes in **50.8%** of months,
 - positive price changes in **29.6%** of months,
 - markup standard deviation of **7.3%**.
- Removing fixed ordering costs (scenario 2) makes ordering continuous – positive orders rise to **100%** – and sharply reduces negative price changes to **26.8%** and markup variability to **3.8%**.
- Imposing symmetric menu costs (scenario 3) has a noticeably smaller effect: negative price changes remain high at **48.5%**, and markup variability falls only to **5.6%**.
- Holding competitors' markups constant (scenario 4) also lowers price reductions and markup variability, showing that exogenous demand variation still matters.
- The paper's bottom-line interpretation is that fixed ordering costs explain about **half** of observed promotions and markup variability; exogenous demand shocks explain much of the rest.

6 Short summary

- **Method.** Build a dynamic model in which a retailer jointly chooses price and order quantity under stockout risk, fixed ordering costs, and menu costs.
- **Identification.** Use reduced-form state dependence, markup movements within ordering spells, and a two-stage structural discrete-choice estimator based on conditional choice probabilities and model-implied exclusion restrictions.
- **Application.** Estimate the model on monthly data for 534 brands from one Spanish supermarket chain between 1990 and 1992.
- **Main result.** Fixed ordering costs are large and explain an important fraction of observed sales promotions and markup variability; menu-cost asymmetry matters, but much less than ordering frictions.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that the interaction between inventories and prices is quantitatively important for understanding retail price dynamics.
- In the data, inventories significantly affect price decisions, especially the decision to run sales promotions.
- The estimated structural model implies that fixed ordering costs, together with stockout risk, explain around half of the observed time variation in markups and around half of the observed sales promotions.

7.2 Economic intuition

- When inventory is high right after an order, the retailer has little stockout risk and can profitably set a low markup to stimulate sales.
- As inventory is depleted, stockout risk rises, effective demand becomes less elastic, and the optimal markup rises.
- Fixed ordering costs make replenishment lumpy. Menu costs then keep many regular prices unchanged between replenishments, producing the observed mix of rigidity and temporary markdowns.
- Wholesaler support for promotions can make price cuts cheaper to implement than price increases, generating asymmetric menu costs.

7.3 Takeaway

This paper is important because it shifts part of the explanation for sales promotions away from purely demand-side stories and toward **inventory management under adjustment costs**. The

main lesson is that one should not interpret temporary price cuts as isolated pricing events. In retail environments with stockout risk and costly ordering, prices and inventories are jointly determined, and much of the observed promotion activity can be a natural consequence of that joint optimization problem.